Lab 6: Linear Statistical Models
In-class Live Coding Challenge

Department of Electrical Engineering, IIT Madras

Demo Date: Apr 8, 2026 during class

1 Regression using Engineered Features

In part B2, it was suggested (optionally) to fit a model that includes the engineered features. Take the three engineered features (water-cement ratio, binder, fine/coarse ratio) and use them in to come up with a new linear regression model. You can additionally use some of the 8 original features as well.

Compare this model with the baseline by providing the same four outputs as in part B2 (coefficient table, R^2 values, Test RMSE, Residual plots on test set).

2 Log-linear model

A log-linear model has the form

$$\exp(c + \sum_i w_i x_i) \tag{1}$$

for features x_i's and weights w_i's. Fit this model as well using the 8 original predictors and provide the four outputs as in the previous part.

Is this a better model than multiple linear regression using the 8 original predictors?

3 Effect of Train-Test Division

We chose 80% of the data to be train data and the rest to be test data. However, the regression model depends on the train-test division.

For 5 different choices of train-test division (such that the train data is still 80%), compare the R^2 values and comment (qualitatively) if the train-test division has a significant impact on the model.